

Ex.No.4b--MACHINE-LEARNING-MODEL-HEART-DISEASE-PREDICTION

AIM

To develop a Heart Disease Prediction model using machine learning classification algorithms and compare the performance of different models using suitable evaluation metrics.

OBJECTIVES

- To understand machine learning classification.
- To analyze a heart disease dataset.
- To identify the input features and target variable.
- To preprocess the dataset.
- To divide the dataset into training and testing data.
- To train different classification models.
- To predict whether a patient has heart disease.
- To evaluate and compare the models.

INTRODUCTION

- Machine Learning enables computers to learn patterns from data and make predictions.
- Classification is a supervised learning technique used to predict categories or classes.
- In this experiment, classification algorithms are used to predict whether a patient is likely to have heart disease.
- The output generally contains two classes: o 0 – No Heart Disease o 1 – Heart Disease

DATASET

The dataset contains medical information about patients. Typical attributes include: Attribute Description Age Age of the patient Sex Gender of the patient Chest Pain Type of chest pain Resting BP Resting blood pressure Cholesterol Cholesterol level Fasting Blood Sugar Blood sugar condition Resting ECG Resting electrocardiogram result Maximum Heart Rate Maximum heart rate achieved Exercise Angina Exercise-induced angina Oldpeak ST depression value Target Presence or absence of heart disease

TARGET VARIABLE

- Target is the dependent variable.
- It indicates whether the patient has heart disease.
- Usually: 0 → No Heart Disease 1 → Heart Disease

Steps

1. Load the dataset.
2. Display the first few records.
3. Check dataset shape.
4. Check data types.

5. Check missing values.
6. Handle missing values if present.
7. Separate features and target.
8. Encode categorical variables if required.
9. Split the dataset into training and testing data.
10. Apply feature scaling where required.

Code

```
import pandas as pd
import numpy as np
```

```
df = pd.read_csv("heart_disease.csv")
```

```
print(df.head())
print(df.shape)
print(df.info())
print(df.isnull().sum())
```

7. SEPARATE INPUT AND OUTPUT
 $X = df.drop("target", axis=1)$
 $y = df["target"]$
 $X \rightarrow$ Patient characteristics.
 $y \rightarrow$ Heart disease prediction.

TRAIN-TEST SPLIT

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
```

80% $\rightarrow$ Training data • 20% $\rightarrow$ Testing data

FEATURE SCALING

```
from sklearn.preprocessing import StandardScaler
```

```
scaler = StandardScaler()
```

```
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

MACHINE LEARNING MODELS

The following models can be compared:

1. Logistic Regression
2. K-Nearest Neighbors
3. Decision Tree
4. Random Forest
5. Support Vector Machine
6. Gradient Boosting

LOGISTIC REGRESSION

```
from sklearn.linear_model import LogisticRegression
```

```
lr = LogisticRegression(max_iter=1000)
```

```
lr.fit(X_train_scaled, y_train)
```

```
y_pred_lr = lr.predict(X_test_scaled)
```

K-NEAREST NEIGHBORS

```
from sklearn.neighbors import KNeighborsClassifier
```

```
knn = KNeighborsClassifier(n_neighbors=5)
```

```
knn.fit(X_train_scaled, y_train)
```

```
y_pred_knn = knn.predict(X_test_scaled)
```

DECISION TREE

```
from sklearn.tree import DecisionTreeClassifier
```

```
dt = DecisionTreeClassifier( random_state=42 )
```

```
dt.fit(X_train, y_train)
```

```
y_pred_dt = dt.predict(X_test)
```

RANDOM FOREST

```
from sklearn.ensemble import RandomForestClassifier
```

```
rf = RandomForestClassifier( n_estimators=100, random_state=42 )
```

```
rf.fit(X_train, y_train)
```

```
y_pred_rf = rf.predict(X_test)
```

SUPPORT VECTOR MACHINE

```
from sklearn.svm import SVC
```

```
svm = SVC(kernel="rbf")
```

```
svm.fit(X_train_scaled, y_train)
```

```
y_pred_svm = svm.predict(X_test_scaled)
```

GRADIENT BOOSTING

```
from sklearn.ensemble import GradientBoostingClassifier
```

```
y_pred_gb = gb.predict(X_test)
```

The models can be evaluated using: • Accuracy • Precision • Recall • F1-score • Confusion Matrix

```
from sklearn.metrics import ( accuracy_score, precision_score, recall_score, f1_score )

models = { "Logistic Regression": y_pred_lr, "KNN": y_pred_knn, "Decision Tree": y_pred_dt,
"Random Forest": y_pred_rf, "SVM": y_pred_svm, "Gradient Boosting": y_pred_gb }

for name, prediction in models.items():

    print(name)
    print("Accuracy :", accuracy_score(y_test, prediction))
    print("Precision:", precision_score(y_test, prediction))
    print("Recall    :", recall_score(y_test, prediction))
    print("F1 Score  :", f1_score(y_test, prediction))
    print()
```

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression				
KNN				
Decision Tree				
Random Forest				
SVM				
Gradient Boosting				

Comparison •

4/5

```
plt.xlabel("Predicted") plt.ylabel("Actual") plt.title("Confusion Matrix - Random Forest")  
  
plt.show()
```

CONCLUSION

Thus, machine learning classification models were successfully applied for heart disease prediction, and their performance was compared using standard classification evaluation metrics.